

APPLIED BUSINESS VALUATION

1. Compare the accounting and the business cash flow statements

- The accounting cash flow statement is prepared for external stakeholders, while the business cash flow statement is an internal tool for management. – primarily used for internal decision making
- Business CF is usually made more frequently

	Accounting	Business
Application	Historical analysis, checking calculations	Predicting, supporting business decisions
Explained quantity	Change in cash	Change in non-operational wealth
Breakdown	Ordinary / Investment / Financial	Operational / Investment / Financing
Aim	Checking on change in cash	Quantifying sub-totals (FCFF, FCFE)

2. List any correction types you may need to use before analyzing the historical statements of a firm

- **Main operation vs non-operative items** (Rented office building)
- **Standard vs Extraordinary items** (Once in a lifetime grant)
- **Operational vs Financing items** (Management fees, exchange gain on interest)

3. List and contrast the DCF models learned. When are those most appropriate to use?

Entity models: FCFF, APV, EVA

Equity models: FCFE, RI

- all of them should have the same result
- **Entity valuation models are usually better:** we usually focus on the value creation process, can be used for Value Based Management, multi-business firms can be analysed, offer more freedom to model financing alternatives.
- FCFF, EVA discount with WACC, FCFE and RI with r_E and APV with r_A
- FCFF – C-F available for all financiers.
LIMITS: 1) shows the value for a given financing structure, 2) financing effects can not be separated >> **APV method** to show financing effects, handy for M&As
3) The seemingly high proportion of terminal value may create the impression that value is created after the explicit period. 4) does not show when the value is created >> **EVA method** to show when the value is created
- FCFE – C-F available to shareholders

Method	To discount	Discount rate	Notes
Total firm DCF	FCFF	WACC	Useful for multiple business units and set D/V ratio firms
Economic profit	EVA / RI	WACC / r_E	Shows when the value will be created.
APV	FCFF and tax shield (TS)	Operative cost of capital (r_A)	Separates the effect of financing, handy for M&As.
Direct equity valuation	FCFE	Required rate on equity (r_E)	Detailed, hectic financial plan available, bottom-up CF, financial service industry

4. What are the most important requirements to consider when estimating the terminal value in a DCF model?

- By the start of the terminal value period all inputs should stabilize
- Ensure that the terminal value reflects the economic conditions and **assumptions used in the explicit forecast period. Consider market growth, intensity of competition, and inflation when choosing ROIC and WACC.** The terminal value should be **consistent with the growth rates, risk factors, and** other assumptions applied during the forecast period.
- Select a reasonable and sustainable terminal growth rate
- **not be over-reliant** about it- a well-constructed DCF model should also place appropriate emphasis on the explicit forecast period to avoid undue sensitivity to terminal value
- $WACC > g_{NOPLAT}$

5. Present the possible methods of estimating the cost of debt.

The cost of debt is the total interest expense owed on a debt. Put simply, the cost of debt is the effective interest rate or the total amount of interest that a company or individual owes on any liabilities, such as bonds and loans. This expense can refer to either the before-tax or after-tax cost of debt. The degree of the cost of debt depends entirely on the borrower's creditworthiness, so higher costs mean the borrower is considered risky.

KEY TAKEAWAYS

- The cost of debt is the effective rate that a company pays on its debt, such as bonds and loans.
- The key difference between the pretax cost of debt and the after-tax cost of debt is the fact that interest expense is tax-deductible.¹
- Debt is one part of a company's capital structure, with the other being equity.
- Calculating the cost of debt involves finding the average interest paid on all of a company's debts.

Effective interest rate $\times (1 - \text{effective tax rate})$

6. What are the value drivers? List them and show the calculation formulas.

NOPLAT (Net Operating Profit Less Adjusted Taxes)

$$\text{NOPLAT} = \text{EBIT} * (1-t)$$

K (Reinvestment rate)

$$K = \Delta \text{IC} / \text{NOPLAT}$$

WACC (Weighted Average Cost of Capital)

$$\text{WACC} = r_A - D/V * r_D * t$$

ROIC (Return on Invested Capital)

$$\text{ROIC} = \text{NOPLAT}(t) / \text{IC}(t-1)$$

g (growth rate)

$$g = \text{NOPLAT}(t+1) / \text{NOPLAT}(t) - 1$$

RONIC (Return on Newly Invested Capital)

$$\text{RONIC} = (\text{NOPLAT}(t) - \text{NOPLAT}(t-1)) / \Delta \text{IC}(t-1)$$

t (effective tax)

$$t (\%) = \text{tax} / \text{Profit before tax}$$

IC (Invested Capital)

$$\text{IC boy} = \text{Debt}(t-1) + \text{Equity}(t-1)$$

$$\Delta \text{IC} = \text{NOPLAT} * K$$

FCFF (Free Cash Flow to Firm)

$$\text{FCFF} = \text{NOPLAT} - \Delta \text{IC}$$

7. How do you have to select the peers for a multiple-based valuation?

Peer companies should **operate in the same or a similar industry**, as different industries may have distinct valuation metrics. Look for peers with **similar business characteristics and revenue drivers**. Take into account the **size and scale of the target company**. Peers should be of comparable size in terms of **revenue, market capitalization, and assets**. Companies **operating in similar regions** may face similar market conditions, regulatory environments, and economic factors. Examine the financial performance of the target company and look for peers with comparable financial metrics, such as revenue growth, profitability, and return on equity. Ensure that the selected peers are **comparable in terms of business model, customer base, product/service offerings**, and other relevant factors. The more similar the companies are, the more reliable the valuation multiples will be. Aim for a diverse set of peers to minimize the impact of idiosyncratic factors on valuation. A mix of competitors, industry leaders, and companies with similar growth stages can provide a well-rounded perspective.

8. What kind of asset valuation methods do you know?

Asset based valuation focuses on the wealth on the assets that the company currently has. We don't really value a business rather than just the assets. We estimate how much money we might be able to redraw if we stop the operation of the firm, liquidate the assets pay back all the liabilities and finish the history of the firm.

1) **Cost Method**

The cost approach values assets based on their **original cost or present replacement cost**. It is often used to value **homes and other tangible assets** for **insurance purposes**. The cost method is the least accurate of all three methods because it **fails to account for depreciation** and other factors that affect

an item's value. The cost method is generally used when the asset's **expected service life is less than one year**.

2) Market Value Method

The market value is the price at which an asset would change hands between a **willing buyer** and a **willing seller**, with neither party under any compulsion to buy or sell, and both having reasonable knowledge of the relevant facts. The market value method can be **used to determine property values** as well as other types of assets like **stocks, bonds, and real estate**. The market value method is based on **current market prices for similar items**. This method is preferred when the **service life of an item exceeds one year** and several vendors are offering similar services at different prices.

3) Income Capitalization Method

This approach estimates the present value of future income generated by an asset by **discounting cash flows** from operations (debt service costs) at an appropriate rate of return on investment (ROI). This is often used to value **real estate properties, businesses, and stock portfolios**. It predicts how much money will be generated by an asset in the next period and then discounts that amount using a discount rate. The sum of all discounted cash flows equals the present value of an investment.

9. Describe how the bargaining range for an M&A transaction set.

The bargaining range in a merger and acquisition (M&A) transaction refers to the range of acceptable prices or terms that both the buyer and the seller are willing to consider. It represents the space within which negotiations can take place, leading to a mutually agreeable deal.

Seller:

Min. price = this is how much the company worth, that's what I would pay (Equity)

+ the transaction fee = final min price

(I add, since in case of the seller, I want at least the equity + legal fees)

Buyer:

Max. price = this is the maximum, how much I will pay, the benefit I get from acquiring the company
– transaction fee = final max price

(In case of the buyer, I am willing to pay E + synergy - the legal and other costs, that I do not want to pay to)